WEEK-11—Amazon LEX

Step 1: Open Amazon Lex

1. Login to AWS Console
2. Search **Amazon Lex**
3. Click **Create bot**

Step 2: Create Bot

1. Choose:
   - **Create a blank bot**
2. Enter:
   - Bot name: HotelBookingBot
3. IAM role → Create new role
4. Remaining default→next
5. Language: English
6. Voice (optional)
7. Click **Done**

Step 3: Create Intent

1. Go to **Intents**
2. Name: BookHotel
3. Click **Create**

Step 4: Add Sample Utterances

Add user sentences (how user talks):

I want to book a hotel

Book a room

Reserve hotel

I need a room

These are **Utterances**

Step 5: Create Slots (User Inputs)

Slots collect user information.

Add Slot

1. Click **Add slot**
2. Fill:
   - Slot name: age
   - Slot type: AMAZON.Number
   - Prompt: *Which city do you want?*

Mark as **Required**

Add IF Conditions (Conditional Logic)

Used to control flow based on slot values

Example Condition:

If {age} < 18

Steps:

1. Go to **Intent → Slots -> under Advance options**
2. Scroll to **Slot capture: success response ->Conditional branching**
3. Click **Add condition**

Example:

Condition:

If {age} < 18

Response:

Your not eligible for hoot booking?

Step 6: Create Slots (User Inputs)

Slots collect user information.

Add Slot

3. Click **Add slot**
4. Fill:
    - Slot name: location
    - Slot type: AMAZON.City
    - Prompt: *Which city do you want?*
5. Mark as **Required**

Repeat for other slots.

1. Click **Add slot**
2. Fill details:
    - **Slot name:** checkin
    - **Slot type:** AMAZON.Date
    - **Prompt:**
      "What is your check-in date?"
3. Mark as **Required**
   (Optional but recommended)
    - Add **Retry prompt**:
      "Please provide a valid date (e.g., 2026-03-25)"
4. Click **Save**

Slot 7: Number of Nights (nights)

Steps:

1. Click **Add slot** again
2. Fill details:
    - **Slot name:** nights
    - **Slot type:** AMAZON.Number

- **Prompt:**
  "How many nights will you stay?"

3. Mark as **Required**
4. Click **Save**

Step 7: Create Custom Slot Type

Save intent

Click on Back to intents

If you want custom values:

1. Go to **Slot Types**
2. Click **Add slot type ->add blank slot type**
3. Name: RoomType

Add values:

Single

Double

Suite

Save slot type

Now use this in slot: go back to intent

- Slot name: RoomType
- Slot type: RoomType

Step 9: Add Response Cards (Card Groups)

Used for buttons (quick replies)

Steps:

1. Go to **slot prompts -> more prompt options**
2. Click **Add -> add card groups**

Example:

Card Title: Select Room Type

click on add Buttons:

- Single → "Single"
- Double → "Double"
- Suite → "Suite"

Step 10: Configure Responses

Initial Response:

Welcome to Hotel Booking! What is your name?

Confirmation:

Do you want to confirm booking in {location} for {nights} nights?

Step 11: Build & Test

1. Click **Build**
2. Use **Test chatbot** panel
3. Try:

Book a hotel

Sample output

User: Book a hotel
Bot: What is your age?
User: 25
Bot: Which city do you want?
User: Hyderabad
Bot: Check-in date?
User: Tomorrow
Bot: Number of nights?
User: 2
Bot: Select room type
User: Double
Bot: Do you want to confirm booking in Hyderabad for 2 nights?
User: Yes
Bot: Booking confirmed.

SCENARIO BASED QUESTIONS

Scenario 1: Bot Not Responding

You created a chatbot, but when you test it, the bot does not respond to user input.

- What could be the possible reason for no response?
- Did you build and publish the bot correctly?
- How would you troubleshoot this issue?

Scenario 2: Incorrect Intent Matching

A user types a valid query, but the chatbot responds with the wrong intent.

- What could cause incorrect intent recognition?

- How do utterances affect intent matching?
- How can you improve accuracy?

Scenario 3: Missing Slot Values

The chatbot asks for user input, but fails to capture required slot values.

- What is a slot in Amazon Lex?
- Why might slot values not be captured properly?
- How would you fix slot configuration issues?

Scenario 4: Fulfillment Not Working

The bot collects all user inputs but does not perform the expected action.

- What is fulfillment in Amazon Lex?
- Could Lambda integration be misconfigured?
- How would you verify the backend processing?

Scenario 5: Bot Not Accessible

You created the bot successfully, but users are unable to access it.

- Did you deploy or publish the bot?
- Could permissions or channel integration be missing?
- How would you make the bot accessible to users?

Scenario 6: Wrong Language Understanding

The chatbot fails to understand user queries written in slightly different formats.

- How does Amazon Lex process natural language?
- Why is training with multiple utterances important?

- How can you improve NLP performance?

Scenario 7: Versioning Issue

You updated the bot, but users still experience the old behavior.

- Did you create a new version of the bot?
- Why is alias configuration important?
- How do you ensure users access the latest version?

Scenario 8: Lambda Permission Error

The chatbot is connected to a Lambda function, but it fails during execution.

- What type of permission error might occur?
- How are IAM roles used in this integration?
- How would you fix the permission issue?

Scenario 9: Bot Gives Default Response

The chatbot always gives a fallback/default response instead of proper answers.

- What is a fallback intent?
- Why does the bot fail to match user input?
- How can you improve intent recognition?

Scenario 10: Multi-turn Conversation Issue

The chatbot fails to continue conversation properly after asking follow-up questions.

- What is dialog management in Amazon Lex?
- How are slots used in multi-turn conversations?
- How can conversation flow be improved?

Scenario 11: Integration with Web Application

A company wants to integrate the chatbot into their website.

- How can Amazon Lex be integrated with web applications?
- What services or APIs are required?
- What challenges might occur during integration?

Scenario 12: Performance Issue

The chatbot responds slowly when many users interact simultaneously.

- What could cause performance delays?
- How can scalability be handled in Amazon Lex?
- What AWS services can support high traffic handling?